

## Lecture 2

### Dedekind vs. Cantor

- Hard to attribute invention of concepts
- “Plagiarism” by Tom Lehrer

$$\mathbb{R} \supset \mathbb{Q}^*$$

- $\mathbb{Q}^*$  is an ordered subfield isomorphic to  $\mathbb{Q}$
- $r \in \mathbb{Q}, r^* = \{s \in \mathbb{Q} : s < r\} \mid \{s \in \mathbb{Q} : s \geq r\}$
- $[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}, a, b \in \mathbb{R}, a < b$

### Theorem

- $\forall [a, b]$  there are infinitely many rationals and infinitely many irrationals contained in  $[a, b]$
- Think of a number line
  - Can draw many dots to symbolize the rationals
  - Between any 2 rationals there are many irrationals
  - Shouldn't think of the numbers as meshed though, more irrationals than rationals
- Let  $a = A \mid A', b = B \mid B'$ 
  - Draw a number line with the 2 cuts  $a$  and  $b$
  - We can choose 2 rationals  $r$  and  $s$  between  $a$  and  $b$ 
    - $B \setminus A$  is nonempty since  $a < b$
    - There must be at least 2 elements.
      - Suppose there is only 1 element in  $B \setminus A$ . Then, we effectively have a cut with 1 extra element, formed by  $A \mid B'$ . This situation would imply that the only rational is  $a$ , meaning  $a = b$ , a contradiction.
- $f : [0, 1] \rightarrow r + t(s - r)$ 
  - $r + t(s - r)$  varies between  $r$  and  $s$
  - This function maps  $[0, 1]$  to  $[r, s]$
  - If  $t$  is rational, then  $r + t(s - r)$  is rational (addition and multiplication with rationals)
  - If  $t$  is irrational, then  $r + t(s - r)$  is irrational
    - Suppose  $r + t(s - r)$  is rational. Then, we can write  $r + t(s - r) = \frac{p}{q}$ . Rearranging (subtraction and division of rationals) gives us  $t = \frac{1}{s-r} \left( \frac{p}{q} - r \right)$ , a rational expression,

### Cartesian Product

- $\mathbb{R} \times \mathbb{R} \times \dots \times \mathbb{R} = \mathbb{R}^m$
- $x \in \mathbb{R}^m, x = (x_1, x_2, \dots, x_m)$

### Inner product

- $\langle x, y \rangle = \sum_{i=1}^m x_i y_i$ 
  - 3 conditions
    - Symmetric
    - Bring constants outside
    - Positive definite ( $x \neq 0 \rightarrow \langle x, x \rangle > 0$ )

### Cauchy-Schwarz Inequality

- $\langle x, y \rangle \leq |x||y|$
- $w = x + ty$
- $|x| = \sqrt{\langle x, x \rangle}$

$$\begin{aligned}\langle w, w \rangle &= \langle x + ty, x + ty \rangle \\ &= \langle x, x \rangle + t\langle y, x \rangle + t\langle x, y \rangle + t^2\langle y, y \rangle\end{aligned}$$

- $\langle w, w \rangle \geq 0$  (positive definite)  $= \langle x, x \rangle + 2t\langle x, y \rangle + t^2\langle y, y \rangle$
- Let  $a = \langle y, y \rangle, b = 2\langle x, y \rangle, c = \langle x, x \rangle$ 
  - We now have a quadratic function  $Q(t) = at^2 + bt + c$
- Since  $\langle w, w \rangle \geq 0, Q(t) \geq 0$ 
  - Since  $Q(t)$  has at most 1 real root,  $b^2 - 4ac \leq 0$
- Substituting values gives us  $4\langle x, y \rangle^2 - 4\langle x, x \rangle\langle y, y \rangle \leq 0$ 
  - Rearranging and simplifying gives us  $\langle x, y \rangle^2 \leq \langle x, x \rangle\langle y, y \rangle$
  - $\langle x, y \rangle \leq \sqrt{\langle x, x \rangle}\sqrt{\langle y, y \rangle}$
  - $\langle x, y \rangle \leq |x| |y|$

### Corollary 1

- $|x + y| \leq |x| + |y|$

$$\begin{aligned}|x + y|^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + 2\langle x, y \rangle + \langle y, y \rangle \\ &\leq |x|^2 + 2|x||y| + |y|^2 \\ &= (|x| + |y|)^2\end{aligned}$$

- Square root both sides to get  $|x + y| \leq |x| + |y|$

### Corollary 2

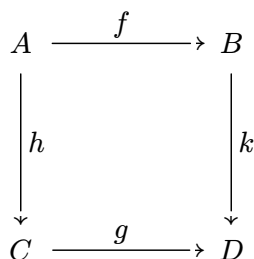
- Points  $x, y, z$  in the plane
- $(z - x) = (z - y) + (y - x)$
- By corollary 1,  $|z - x| \leq |z - y| + |y - x|$

## Definitions and Vocab

- Distance
  - $d(x, y) = |x - y|$
- Ball
  - For a point  $p \in \mathbb{R}^m$  and a radius  $r > 0$
  - $B(r, p) = \text{ball of } r \text{ at point } p = \{x \in \mathbb{R}^m : |x - p| \leq r\}$
  - Closed ball
  - Isometric transformations, still get the same ball
- Sphere
  - For a point  $p \in \mathbb{R}^m$  and a radius  $r > 0$
  - Surface relative to solid object (ball), so dimension is 1 lower.
  - $r = S^{m-1} = \{x \in \mathbb{R}^m : |x - p| = r\}$
- When  $m = 2$ 
  - Ball is a disc of radius  $r$  at  $p$
  - Sphere becomes circle
  - Be careful of distinction here
- Box in  $\mathbb{R}^m$ 
  - $[a_1, b_1] \times \dots \times [a_m, b_m]$
  - $m = 2 \rightarrow$  filled in square
  - Unit box when all intervals are unit intervals
  - Changes with transformations in parallelepiped

- ▶ Rational, irrational vectors (all terms are rational/irrational), box contains both by earlier theorem
- Continuous functions
  - ▶  $C^0([0, 1], \mathbb{R})$ 
    - Continuous functions defined on  $[0, 1]$  of real numbers
  - ▶  $f, g \in C^0([0, 1], \mathbb{R})$
  - ▶  $\langle f, g \rangle = \int_0^1 f(t)g(t)dt$  is an inner product
    - Fulfills all 3 conditions: symmetric, positive definite (squaring  $f > 0$ ), can pull out scalars
    - Since inner product is defined, Cauchy-Schwarz holds
  - ▶  $\int fg \leq \sqrt{\int f^2} \sqrt{\int g^2}$
- More about functions
  - ▶ A, B are sets
  - ▶  $f : A \rightarrow B$ 
    - Function is the thing that transforms/sends elements of A to B
  - ▶
 

| Concept                                      | Old              | Bourbaki   | Structure    |
|----------------------------------------------|------------------|------------|--------------|
| $f(a) = f(b) \rightarrow a = b$              | one-to-one / 1:1 | injective  | monomorphism |
| $\forall b \in B \exists a \in A (f(a) = b)$ | onto             | surjective | epimorphism  |
| both of the above                            | 1:1 & onto       | bijective  | isomorphism  |
  - ▶ Bourbaki column will be used in this class
  - ▶ History of Bourbaki
    - French mathematicians after WW1
    - Everything is abstract, creating encyclopedia of math
      - No matrices, just linear maps
      - Anonymous group
  - ▶  $A \xrightarrow{f} B \xrightarrow{g} C$ 
    - $g \circ f$
    - $g \circ f(a)$ , not  $f \circ g(a)$ ;  $a$  is not necessarily in  $B$
    - composition of injective functions preserves injection
      - Same for surjection and bijection
  - ▶ Commutative Diagrams



- Commutative if  $k \circ f = g \circ h$
- Start with A, follow diagram, 2 paths to same answer
- Examples
  - Partial derivatives
  - Bolts
    - ▶ Attaching hexagonal part and threading can be done in either order; commutative
    - ▶ Electroplating can't; not commutative